



وزارة الاتصالات
وتكنولوجيا المعلومات



CAPSTONE PROJECT

Nano Degree Program

SkillBridge

A Data-Driven System for Analyzing Freelance Market Trends,
Skills Demand, and Career Opportunities

Graduation Project Report

by

Rasha Wael Ramadan

Alaa Hussein Mohamed

Esraa Metwally Abouelnasr

Fatma Ibrahim Morsy

Submitted in Partial Fulfillment of the Requirements for the
Nano Degree Program in Data Analytics

Supervisor

Prof. Dr. Rania El Gohary

Track Head

Dr. Ahmed Zakaria

Academic Team

Dr. Yasser Ali

Data Analytics Track

September 2026
Egypt



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*From
Data
to a Brighter
Future*

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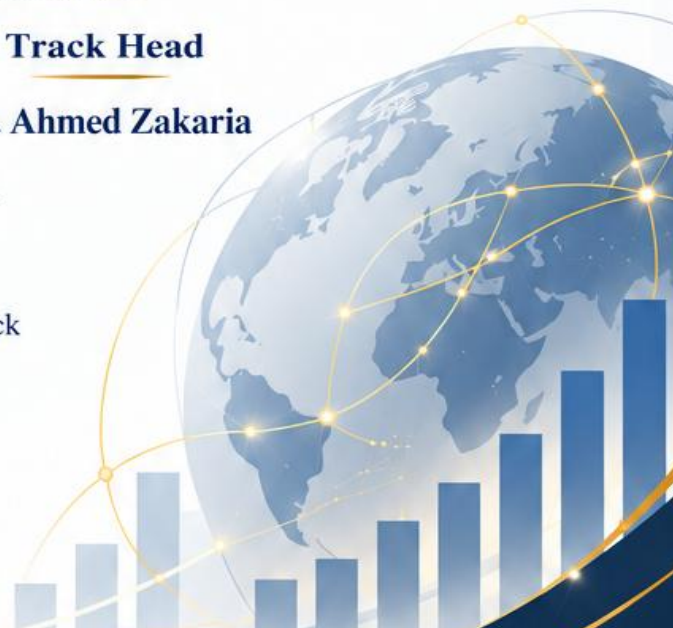
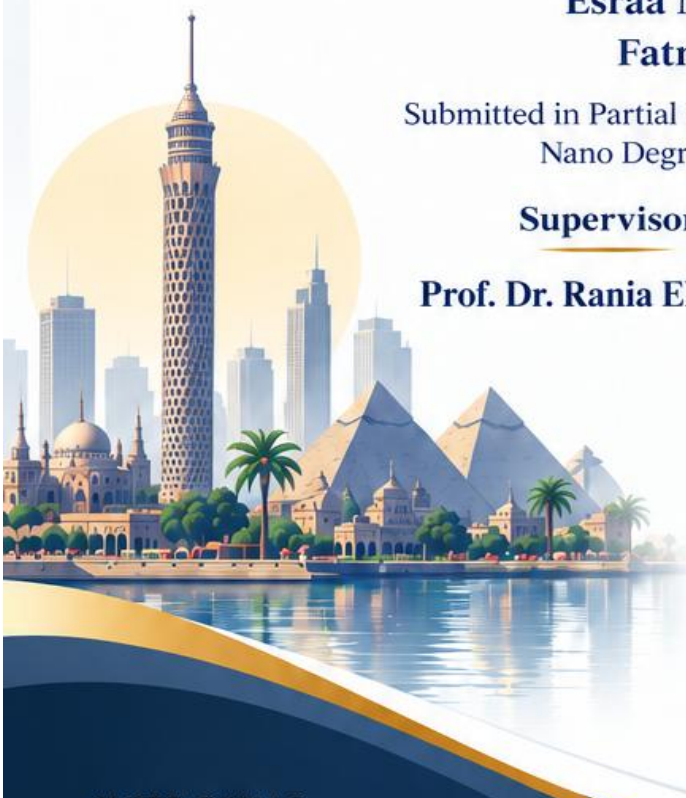
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-  Analyze
-  Explore
-  Empower
-  Grow





CAPSTONE PROJECT PROPOSAL

Project Title	SkillBridge
Track	Data Analytics Track
Job Profile	Junior Data Analyst
Program Name	Nano Degree
Team Name	LexiTech
Submitted By	Alaa Hussien Mohamed, Rasha Wael Ramadan, Fatma Ibrahim Morsy, Esraa Metwally Abo Elnasr
Supervision Committee	Dr. Ahmed Zakaria (Track Head) Prof. Dr. Rania ELgohary (Supervisor) Dr. Yasser Ali (Academic Team)
Submission Date	September 2026

Declaration and Certificate of Originality

We, the undersigned team members, declare that this capstone project proposal is our original work and has been prepared for academic evaluation purposes. Any external datasets, tools, libraries, frameworks, or resources used in the project will be clearly acknowledged within this final project documentation according to the committee requirements.

We confirm that the proposed project idea, implementation plan, and expected deliverables are developed by the team and that the project will follow ethical, legal, and professional standards related to data, security, software development, and technology usage.

Student Name	ID	Signature	Date
Alaa Hussien Mohamed			September 2026
Fatma Ibrahim Morsy			September 2026
Rasha Wael Ramadan			September 2026
Esraa Metwally Abo Elnasr			September 2026

Dedication

We dedicate this project to our families, whose endless support, patience, and encouragement have been a constant source of strength throughout our learning journey.

We also dedicate this work to our professors, mentors, and instructors, whose guidance and knowledge inspired us to transform what we learned into a practical data analytics project with meaningful real-world value.

This project is dedicated to every ambitious Egyptian youth seeking to develop their digital skills, enter the freelance market, and build a sustainable career in an increasingly data-driven and globally connected world.

Finally, we dedicate SkillBridge to the vision of a more skilled and opportunity-driven future for Egypt, where data can help bridge the gap between learning, market demand, and career opportunities.

We extend our sincere appreciation to the Egyptian Military Academy, the Ministry of Communications and Information Technology (MCIT), and the DiGiLiANS initiative for providing this valuable learning opportunity and supporting the development of Egypt's future digital talent.

Acknowledgement

The team would like to express its sincere gratitude to the DiGiLiANS supervision committee, track head, and academic supervisors for their continuous guidance, invaluable insights, and unwavering support throughout this project. Special thanks to our mentors and technical advisors, whose constructive feedback shaped our understanding of real-world data analytics, business intelligence, and labor market dynamics.

We also acknowledge the open-source tools, platforms, and vibrant technical communities behind Python, Pandas, Power BI, and Kaggle that made this data ingestion, ETL modeling, and interactive visualization framework possible.

Abstract

This project proposes the development of an integrated Data Analytics and Business Intelligence Framework, named SkillBridge, to address the structural mismatch between pre-university educational outcomes and the dynamic skill demands of the global freelance market. The proposed system aims to support key national stakeholders—including the Ministry of Communications and Information Technology (MCIT), the Ministry of Education and Technical Education, the Ministry of Higher Education and Scientific Research, and private EdTech/programming academies—by providing data-driven market transparency, wage rate benchmarking, skill-gap analysis, and automated skill-match job aggregation.

The solution utilizes a robust 4-tier technical pipeline leveraging Python (Pandas, NumPy) for web scraping and ETL processing, Jupyter Notebook for data normalization, a relational Star Schema data model, and Power BI Desktop for dynamic DAX-driven reporting. The expected output is a high-performance suite of interactive dashboards paired with an automated job matching mechanism that improves educational ROI, optimizes national IT training investments, reduces freelancer proposal friction, and accelerates digital talent alignment with global remote work opportunities.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DE	Data Engineering
BD	Big Data
API	Application Programming Interface
EdTech	Educational Technology
EDA	Exploratory Data Analytics
MVP	Minimum Viable Product
MCIT	Ministry of Communications and Information Technology
K-12	Kindergarten through 12th Grade
CatBoost	Categorical Boosting
ETL	Extract, Transform, Load
DAX	Data Analysis Expressions
ROI	Return on Investment
MAE	Mean Absolute Error
R^2	Coefficient of Determination

List of Symbols

Symbol	Meaning / Description
$[x]$	[Input variable / feature / signal]
$[y]$	[Output label / target variable]
$[t]$	[Time step / timestamp, if applicable]
$[N]$	[Number of records / users / events]

1. Executive Summary

• Problem Statement & Core Vision

The rapid growth of digital platforms and remote work has transformed freelancing into an increasingly important source of employment and income. However, the large volume of available jobs, skills, freelancers, platforms, and market information makes it difficult for individuals to identify the most suitable opportunities, understand market demand, evaluate competition, and determine appropriate pricing for their services. SkillBridge was developed as an integrated Data Analytics, Business Intelligence, and Machine Learning solution designed to transform complex freelance-market data into meaningful and actionable insights. The project aims to bridge the gap between talent and market demand by analyzing the relationships among freelance jobs, required skills, freelancers, clients, platforms, geographical markets, compensation levels, and competition.

• End-to-End Analytics Workflow & BI Modeling

The project follows an end-to-end data analytics workflow that includes data cleaning, transformation, integration, validation, and analytical modeling. Data related to freelance jobs, skills, freelancers, clients, countries, platforms, and other relevant entities is organized into a structured data model. Using Power BI and DAX, a comprehensive set of Key Performance Indicators (KPIs) and analytical measures was developed to evaluate job demand, skill demand and supply, hiring activity, applicant competition, hourly rates, project budgets, market growth, and opportunity potential.

• Interactive Intelligence Dashboards

Interactive Power BI dashboards provide different analytical perspectives on the freelance market. The solution includes **Freelance Market Intelligence** for monitoring overall market trends and performance, **Skill Gap Intelligence** for comparing skill demand with available talent, and **Diamond Skills Finder** for identifying potentially valuable skills based on market demand, supply, competition, and earning potential.

• Predictive ML Salary Engine & Performance

In addition to Business Intelligence analytics, SkillBridge incorporates a Machine Learning-based **Salary Predictor**. The predictive component estimates the expected hourly rate of a freelance job based on available job characteristics, including experience level, category, country, budget type, project duration, platform type, and skill-related information. The Machine Learning component was developed using a CatBoost Regressor, with Linear Regression used as a baseline model for performance comparison. The final CatBoost model was trained using **109,470 valid job records** and achieved an R^2 score of **0.682**, a **Mean Absolute Error (MAE) of \$2.64 per hour**, and an **RMSE of 5.40**. Five-Fold Cross-Validation was also applied, producing an average MAE of approximately \$2.64 with low variation across folds, indicating stable predictive performance.

- **Methodological Integrity & Data Leakage Prevention**

To improve the reliability of the predictive process, post-publication variables such as *Applicants Count*, *Proposals Count*, *Hire Rate*, and *Job Status* were excluded from the model features because they would not be available when predicting the expected rate of a newly defined job. This approach helps prevent Data Leakage and makes the model more appropriate for practical prediction scenarios.

- **Stakeholder Impact & Strategic Alignment**

SkillBridge is intended to support students, aspiring and existing freelancers, training providers, educational institutions, career advisors, and stakeholders involved in digital workforce development. By integrating descriptive and diagnostic analytics with predictive Machine Learning capabilities, the project provides a more comprehensive understanding of the freelance market and supports evidence-based decisions related to skills, pricing, learning priorities, and career opportunities.

- **Unified Decision-Support Architecture**

The final solution therefore combines Business Intelligence dashboards, market and skill-gap analytics, opportunity intelligence, and Machine Learning-based salary prediction within a unified decision-support framework. Through this integration, SkillBridge transforms freelance-market data into practical intelligence and supports its central vision of bridging the gap between learning, skills, market demand, and earning opportunities.

2. Problem Statement

The rapid growth of freelancing has created significant opportunities for digital employment; however, the market has also become increasingly complex. Freelancers and individuals planning to enter the market often face difficulty identifying which skills are truly in demand, where competition is lower, which opportunities provide better earning potential, and what hourly rate is appropriate for a specific job. Freelance-market information is also distributed across jobs, skills, freelancers, clients, platforms, countries, and categories, making it difficult to convert raw data into clear career decisions.

A major challenge is the mismatch between skill demand and skill supply. A skill may be widely available among freelancers while having relatively limited job demand, while another skill may have strong demand but insufficient available talent. Evaluating skills based only on popularity can therefore lead to poor learning and career decisions. SkillBridge addresses this problem by measuring Skill Demand, Skill Supply, Skill Gap, Demand-Supply Ratio, and Skills Readiness Gap, allowing users to identify areas where market needs and available talent are not well aligned.

Competition represents another challenge. A high number of freelance jobs does not necessarily indicate a strong opportunity when each job attracts many applicants. SkillBridge therefore combines job demand with indicators such as Average Applicants per Job, Hire Rate, Competition Index, and Opportunity Ratio to provide a more realistic assessment of market opportunities.

Pricing is another significant problem in the freelance market. The expected value of a job can vary according to experience level, category, country, project duration, budget type, platform, and required skills. Without analytical support, freelancers may find it difficult to determine a reasonable market rate. To address this issue, SkillBridge includes a Machine Learning–based Salary Predictor that estimates the expected hourly rate of a freelance job using its available characteristics.

The predictive component uses a CatBoost Regressor, selected after comparison with a Linear Regression baseline. The final model was trained on 109,470 valid job records and achieved an R^2 of 0.682, MAE of \$2.64 per hour, and RMSE of 5.40, providing a quantitative predictive layer alongside the Business Intelligence analysis.

To solve these interconnected problems, SkillBridge integrates Data Analytics, Business Intelligence, and Machine Learning within a unified decision-support solution. Interactive Power BI dashboards analyze market trends, skills gaps, competition, financial indicators, platforms, and opportunities, while the Diamond Skills Finder supports the identification of promising skills based on multiple market factors. The Salary Predictor extends the solution from descriptive and diagnostic analysis toward predictive analytics.

Therefore, the central problem addressed by SkillBridge is not simply the availability of freelance jobs, but the difficulty of transforming fragmented market data into reliable decisions about what to learn, where opportunities exist, how competitive the market is, and what a job may reasonably be worth. SkillBridge addresses this gap by connecting learning, skills, market demand, opportunity intelligence, and earning potential within one integrated analytical framework.

3. Project Objectives

The main objective of SkillBridge is to develop an integrated Data Analytics, Business Intelligence, and Machine Learning solution that transforms freelance-market data into actionable insights and supports informed decisions related to skills, opportunities, competition, pricing, and career development.

The project aims to analyze freelance market trends by examining job demand, categories, platforms, experience levels, countries, hiring activity, and changes in opportunities over time. It also aims to measure Skill Demand and Skill Supply and identify potential market gaps using analytical indicators such as Skill Gap, Demand-Supply Ratio, and Skills Readiness Gap.

Another objective is to evaluate market competition through indicators such as Average Applicants per Job, Hire Rate, Competition Index, and Opportunity Ratio, providing a more realistic understanding of opportunities rather than relying only on the number of available jobs. SkillBridge also analyzes financial indicators, including hourly rates, project budgets, and market rate gaps, to provide insight into earning potential across different market segments.

The project aims to develop interactive Power BI dashboards that present KPIs, market trends, skill gaps, competition levels, financial indicators, and opportunity insights in a clear and user-friendly format. The Diamond Skills Finder further supports the identification of potentially valuable skills by considering multiple factors such as demand, supply, competition, and earning potential.

A key predictive objective is to develop a Machine Learning–based Salary Predictor capable of estimating the expected hourly rate of a freelance job from its available characteristics. The implemented model uses CatBoost Regression and considers factors such as experience level, category, country, budget type, project duration, platform type, and skill-related information.

The project also aims to evaluate the reliability of the predictive model using appropriate Machine Learning metrics and validation techniques. The final CatBoost model achieved an R^2 of 0.682, MAE of \$2.64 per hour, and RMSE of 5.40, with Five-Fold Cross-Validation used to assess model stability.

Ultimately, SkillBridge aims to create a unified decision-support framework that connects market intelligence, skill-gap analysis, opportunity discovery, salary prediction, and career development, helping bridge the gap between what individuals learn, what the freelance market demands, and where sustainable earning opportunities exist.

4. Project Scope

The scope of SkillBridge covers the development of an integrated Data Analytics, Business Intelligence, and Machine Learning solution for analyzing the freelance market and supporting data-driven decisions related to skills, competition, opportunities, and expected hourly rates.

The project includes the preparation, cleaning, transformation, integration, and modeling of data related to freelance jobs, skills, freelancers, clients, platforms, countries, categories, and educational initiatives. These data sources are connected within a structured analytical model to provide a comprehensive view of the freelance market.

The Business Intelligence scope includes analyzing job demand, market trends, skill demand and supply, skill gaps, applicant competition, hiring activity, experience levels, platforms, geographical markets, hourly rates, and project budgets. Analytical measures such as Skill Gap, Demand-Supply Ratio, Competition Index, Opportunity Ratio, and market growth indicators are used to evaluate different aspects of the market.

The project also includes the development of interactive Power BI dashboards that enable users to explore KPIs, market trends, skill gaps, financial indicators, competition levels, and potential opportunities through dynamic visualizations and filtering capabilities. Features such as Freelance Market Intelligence, Skill Gap Intelligence, Diamond Skills Finder, and Opportunity Intelligence provide different perspectives on the analyzed data.

The Machine Learning scope includes the development of a Salary Predictor that estimates the expected hourly rate of a freelance job based on information available about the job. The model considers factors such as experience level, category, country, budget type, project duration, platform type, and skill-related characteristics. The implemented solution uses a CatBoost Regressor, with Linear Regression serving as a baseline for performance comparison.

The predictive model was developed using 109,470 valid job records and evaluated using MAE, RMSE, R^2 , and Five-Fold Cross-Validation. The final CatBoost model achieved an R^2 of 0.682, MAE of \$2.64 per hour, and RMSE of 5.40.

The current scope of SkillBridge is focused on market analytics, Business Intelligence, skill-gap analysis, opportunity evaluation, and hourly-rate prediction. It does not operate as a complete freelance marketplace, recruitment platform, or job-application system. In addition, the Salary Predictor estimates the expected market rate of a job rather than the personal rate of an individual freelancer, as the current dataset does not contain complete real-world candidate profiles and CV information.

Within these boundaries, SkillBridge provides an integrated decision-support framework that connects market demand, talent supply, competition, financial potential, Machine Learning prediction, and career opportunities, supporting the project's overall goal of bridging the gap between talent and market demand.

5. Target Users and Stakeholders

SkillBridge is primarily designed to support governmental, educational, and training stakeholders responsible for developing digital skills and preparing individuals for the future labor market. By combining freelance market intelligence, skill-gap analysis, opportunity indicators, and Machine Learning–based salary prediction, the project provides evidence-based insights that can support training priorities, curriculum development, digital workforce planning, and strategic decision-making.

Stakeholder	Role / Benefit
Ministry of Communications and Information Technology (MCIT)	Can use SkillBridge insights to identify high-demand digital skills, emerging freelance opportunities, skill gaps, and earning trends, supporting the planning and evaluation of national digital-skills and freelancing initiatives
Ministry of Education and Technical Education	Can benefit from market-driven skill insights to better align technical and pre-university education with emerging digital skills and future freelance-market requirements.
Ministry of Higher Education and Scientific Research	Can use labor-market and skill-gap intelligence to support curriculum development, career-readiness programs, and stronger alignment between university learning outcomes and evolving digital-market needs.
Private Programming and Technology Training Providers	Programming academies, training centers, and EdTech companies can use SkillBridge to identify high-demand and high-potential skills, helping them design and update courses according to actual market demand rather than assumptions.
National Digital Skills & Workforce Development Programs	National initiatives focused on digital transformation, youth employment, and workforce development can use SkillBridge as a decision-support tool to identify priority skills, evaluate market opportunities, and direct training investments toward areas with stronger employment and earning potential.
Students, Graduates, and Freelancers	As indirect beneficiaries, individuals can use SkillBridge insights to understand market demand, identify promising skills, evaluate competition, explore career opportunities, and estimate expected hourly rates through the Machine Learning Salary Predictor.

6. Proposed Solution

SkillBridge proposes an integrated Data Analytics, Business Intelligence, and Machine Learning solution designed to bridge the gap between education, digital skills development, and the actual requirements of the freelance market. The solution transforms fragmented market data into actionable intelligence that can support governmental institutions, educational organizations, training providers, and individuals in making evidence-based decisions.

The solution integrates data related to freelance jobs, skills, freelancers, clients, platforms, countries, job categories, and educational initiatives. The data is cleaned, transformed, validated, and organized within a structured analytical model, allowing different dimensions of the freelance market to be analyzed together rather than independently.

Using Power BI and DAX, SkillBridge provides interactive dashboards and Key Performance Indicators (KPIs) for monitoring job demand, skill demand and supply, skill gaps, competition levels, hiring activity, hourly rates, project budgets, platforms, geographical markets, and market growth. Indicators such as Skill Gap, Demand-Supply Ratio, Competition Index, and Opportunity Ratio help transform market activity into measurable opportunity intelligence.

The solution includes Skill Gap Intelligence to identify differences between skills required by the market and available talent, while the Diamond Skills Finder highlights potentially valuable skills by considering demand, supply, competition, and earning potential. These capabilities can help institutions such as MCIT, the Ministry of Education and Technical Education, the Ministry of Higher Education and Scientific Research, and private technology training providers identify priority digital skills and better align training and educational programs with market requirements.

SkillBridge also incorporates a Machine Learning–based Salary Predictor that estimates the expected hourly rate of a freelance job using characteristics such as experience level, category, country, budget type, project duration, platform type, and skill-related information. The implemented model uses CatBoost Regression and was trained on 109,470 valid job records. The final model achieved an R^2 of 0.682, MAE of \$2.64 per hour, and RMSE of 5.40, adding a predictive analytics layer to the Business Intelligence solution.

At the strategic level, SkillBridge can function as a decision-support framework for digital workforce development. Market and skill-gap intelligence can help stakeholders identify where training investment may be needed, which digital skills show stronger market potential, and how educational and training priorities can be better connected with evolving freelance-market opportunities.

Therefore, the proposed solution creates an analytical bridge between Education → Skills → Market Demand → Opportunities → Earning Potential. By combining Business Intelligence with Machine Learning, SkillBridge moves beyond traditional market reporting toward a data-driven system that can support skills planning, workforce development, and future-ready career pathways.

7. System Features

SkillBridge includes a set of analytical and interactive features designed to transform free-lance market data into clear insights for skills analysis, opportunity evaluation, competition assessment, and career decision-making.

Feature	Description
Interactive User Interface	An intuitive Power BI interface offering interactive visuals, dynamic cross-filtering, and comparative market analysis for key stakeholders.
Data Preparation and Integration	Cleans, transforms, validates, and integrates data related to jobs, skills, freelancers, clients, platforms, countries, categories, and educational initiatives.
Processing and Data Modeling Module	Organizes the integrated data into a structured analytical model and establishes relationships between the main project entities to support accurate analysis.
Market Analytics Module	Analyzes job demand, market trends, experience levels, platforms, hiring activity, competition, hourly rates, project budgets, and geographical market patterns.
Skill Gap Intelligence	Compares Skill Demand and Skill Supply and calculates indicators such as Skill Gap, Demand-Supply Ratio, and Skills Readiness Gap to identify areas where market demand exceeds available talent.
Diamond Skills Finder	Identifies high-potential digital skills by combining demand, supply, competition, and earning potential to support training priorities and skills-development decisions.
Machine Learning – Salary Predictor	Uses a CatBoost Regression model to estimate the expected hourly rate of freelance jobs based on characteristics such as experience level, category, country, budget type, project duration, platform type, and skill-related information.
Opportunity Intelligence Module	Combines market demand, competition, hiring activity, and financial indicators to support the identification and comparison of promising freelance opportunities
Interactive Dashboards	Displays KPIs, market trends, skill gaps, competition indicators, financial insights, platform performance, and opportunity analysis through Power BI dashboards.
Strategic Workforce Insights	Delivers data-driven insights to align national digital curricula and training programs with real-time free-lance market demands.
Reporting and Export	Enables dashboard views, KPIs, and analytical findings to be presented and exported for reporting, evaluation, and decision-support purposes.

8. Technical Approach

SkillBridge follows an end-to-end technical approach that combines Data Preparation, Data Modeling, Business Intelligence, and Machine Learning. The workflow starts with integrating multiple freelance-market datasets, continues through analytical modeling and dashboard development, and ends with predictive analysis using a Machine Learning–based Salary Predictor.

8.1 Input / Data Sources

The project uses structured tabular data related to the freelance market. The main data sources include:

- **Freelance Jobs Data:** Contains job information such as job category, experience level, budget type, project duration, hourly rate, client country, platform, and other job-related attributes.
- **Skills Data:** Contains information about digital skills, skill categories, and the skills required for freelance jobs.
- **Freelancers Data:** Includes freelancer-related information such as experience, hourly rates, ratings, completed jobs, and available skills.
- **Clients Data:** Contains information about clients, including country, ratings, previous hires, and spending activity.
- **Countries Data:** Provides geographical information such as country, region, currency, and Cost Index.
- **Platforms Data:** Contains information about freelance platforms and their platform types.
- **Category Data:** Provides the classification and names of freelance job categories.
- **Educational Initiatives Data:** Includes information related to training and educational initiatives that can be compared with freelance-market indicators.
- **Job-Skills Relationship Data:** Connects freelance jobs with the skills required for each job and their required skill levels.

For the Machine Learning Salary Predictor, five main data sources were integrated: freelance jobs, category data, countries, job-skills relationships, and platforms. The freelance jobs table contained 110,550 jobs, while the job-skills table contained 380,207 records. After constructing the prediction target and filtering invalid or extreme values, 109,470 valid job records were retained for model development.

The data used in SkillBridge is primarily structured tabular data. It is cleaned, transformed, validated, and integrated before being used for Business Intelligence analysis, interactive Power BI dashboards, and Machine Learning modeling.

8.2 Data / Input Preparation

- The data preparation stage in SkillBridge focuses on ensuring that all datasets are accurate, consistent, and suitable for both Business Intelligence analysis and Machine Learning modeling. The preparation process includes data cleaning, transformation, integration, validation, and feature engineering.
- The main preparation steps include data cleaning to identify and handle missing, invalid, duplicated, and inconsistent values; data transformation to standardize columns, correct data types, and prepare values for analysis; and data integration to connect related datasets such as freelance jobs, categories, countries, platforms, job skills, freelancers, clients, and supporting tables.
- Data validation is also performed to ensure logical consistency and reduce errors before the data is used in dashboards or predictive models. Feature engineering is applied where required to create analytical and predictive variables. For the Salary Predictor, a unified target variable called `Estimated_Hourly_Rate` was created so that hourly and fixed-price jobs could be analyzed using the same measurement scale.
- To prevent Data Leakage, variables such as `Applicants_Count`, `Proposals_Count`, `Hire_Rate`, and `Job_Status` were excluded from the Machine Learning features because these values are only available after a job has been published or hiring activity has occurred.
- Invalid and extreme target values were also filtered before model training, resulting in 109,470 valid job records used for the Machine Learning stage.
- The output of this stage is a clean, validated, and integrated dataset that can be used reliably for Data Analytics, Power BI reporting, and Machine Learning.

8.3 Core Development Method by Track

Track	Possible Technical Approach
Data Analytics / Business Intelligence	Data integration, analytical modeling, KPI development, DAX measures, market trend analysis, skill-gap analysis, competition analysis, financial analysis, and interactive Power BI dashboard development.
AI / Machine Learning	Regression model development for hourly-rate prediction, feature engineering, Data Leakage prevention, model training, model comparison, Train/Test evaluation, Five-Fold Cross-Validation, and performance assessment using MAE, RMSE, and R ² . The final model is CatBoost Regressor, with Linear Regression used as the baseline model.
Software Development / API Integration	Deployment and integration of the trained Machine Learning model using FastAPI, Uvicorn, Pydantic, and Joblib to enable Salary Predictor requests within the SkillBridge solution.

The following figure illustrates the Power BI data model developed for SkillBridge and the relationships between the main project tables.

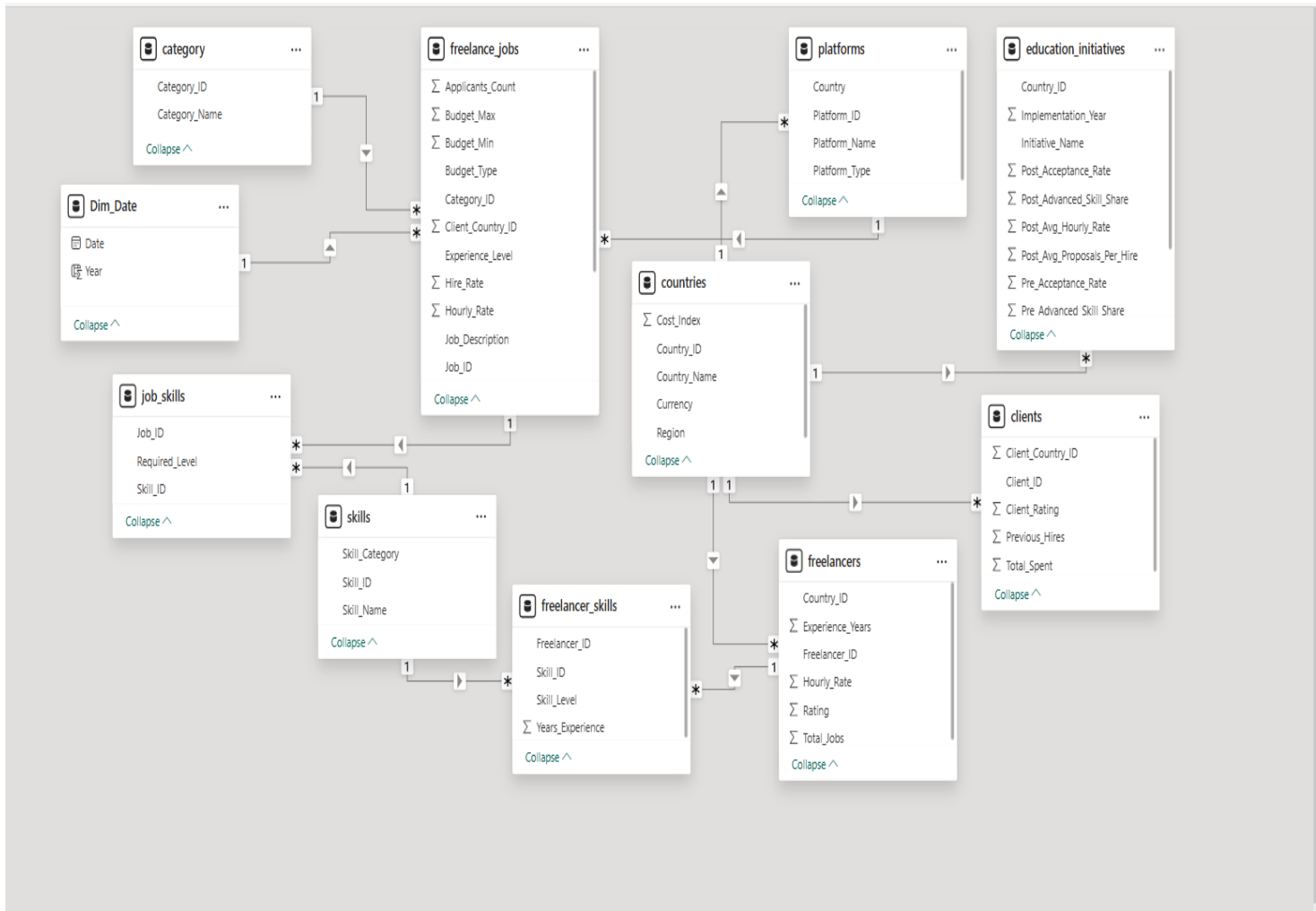


Figure 1 SkillBridge Power BI Data Model and Table Relationships

9. Proposed System Architecture

SkillBridge follows a layered system architecture that integrates freelance market data, data preparation, Business Intelligence, Machine Learning, and API services within a unified analytical framework. The architecture transforms raw data into interactive insights and predictive outputs that support skill-gap analysis, market opportunity evaluation, workforce planning, and data-driven decision-making.

Freelance Jobs / Skills / Freelancers / Clients / Countries / Platforms / Categories / Educational Initiatives
Data Cleaning / Transformation / Integration / Validation / Feature Engineering
Data Modeling / DAX Measures / KPIs / Skill Gap Analysis / Competition Analysis / Opportunity Analysis
CatBoost Machine Learning Model / Salary & Hourly-Rate Prediction / Model Evaluation / Feature Importance
FastAPI / Pydantic / Joblib / Uvicorn / Model Serving & Prediction Requests
Power BI Dashboards / Salary Predictor / Market Intelligence / Skill Gap Intelligence / Diamond Skills Finder / Strategic Decision Support

10. Tools and Technologies

The following tools and technologies were selected based on their relevance to the Skill-Bridge data analytics, Business Intelligence, Machine Learning, and API integration workflow.

Category	Examples
Programming Languages	Python, DAX
AI / ML Libraries	Scikit-learn, CatBoost, Pandas, NumPy, Joblib
Data Preparation & Modeling	Power Query, Power BI Data Model
Backend / API Integration	FastAPI, Uvicorn, Pydantic
Visualization & Business Intelligence	Microsoft Power BI
Data Storage / Input Files	Microsoft Excel, CSV Files
Development Environment	Jupyter Notebook, Visual Studio Code
Version Control	Git, GitHub
Visualization	Power BI, Tableau, Plotly, Matplotlib

11. Data / System Requirements

SkillBridge requires reliable market data, suitable analytical and Machine Learning software, adequate computing resources, and clearly defined functional and non-functional requirements to support accurate analysis, prediction, and decision-making.

Requirement Type	Description
Data Requirements	Structured freelance market data covering jobs, skills, freelancers, clients, countries, platforms, categories, educational initiatives, and job-skill relationships. The data should be complete, consistent, cleaned, and suitable for Business Intelligence and Machine Learning analysis.
Functional Requirements	The system should analyze market trends, calculate KPIs, identify skill demand and supply gaps, evaluate competition and opportunities, provide interactive Power BI dashboards, and generate hourly-rate predictions through the Machine Learning Salary Predictor.
Non-Functional Requirements	The solution should provide good performance, usability, reliability, maintainability, clear visualizations, and consistent analytical results when processing large datasets.
Hardware Requirements	A standard computer capable of running Microsoft Power BI Desktop, Python, and Machine Learning workloads, with sufficient RAM, processing power, and storage for data analysis and model training.
Software Requirements	Microsoft Power BI Desktop, Power Query, Python, Pandas, NumPy, Scikit-learn, CatBoost, Joblib, FastAPI, Uvicorn, Pydantic, Jupyter Notebook / Visual Studio Code, Microsoft Excel, Git, and GitHub.
Security Requirements	Data should be handled securely with controlled access to project files and models. The solution should avoid exposing sensitive or unnecessary personal information and should maintain the integrity of analytical and prediction outputs.

12. Expected Deliverables

The expected deliverables of SkillBridge include the analytical, predictive, technical, and documentation outputs developed throughout the project lifecycle.

- **Project Proposal Document**
- **Cleaned and Integrated Freelance Market Dataset**
- **Power BI Data Model**
- **Interactive Power BI Dashboards**
- **DAX Measures and KPI Framework**
- **Skill Gap Intelligence Analysis**
- **Diamond Skills Finder**
- **Freelance Opportunity Intelligence**
- **Machine Learning Salary Predictor**
- **Trained CatBoost Regression Model**
- **Machine Learning Evaluation Report**
- **FastAPI Model Integration**
- **Source Code Repository**
- **Final Project Documentation**
- **Final Presentation**
- **Demo Video**

13. Evaluation and Testing

SkillBridge is evaluated through a combination of data quality assessment, Business Intelligence validation, Machine Learning performance evaluation, and API testing to ensure that the analytical and predictive outputs are accurate, reliable, and suitable for decision-support purposes.

Project Type	Possible Evaluation Methods
Data Analytics / Business Intelligence	Data quality checks, validation of calculated measures and KPIs, relationship verification, dashboard consistency testing, filter and slicer testing, and visual accuracy assessment.
Machine Learning – Salary Predictor	Model comparison, Train/Test evaluation, Five-Fold Cross-Validation, MAE, RMSE, R^2 , error analysis, feature importance analysis, and Data Leakage prevention.
API Integration	Functional API testing, input validation, prediction-response testing, error handling, and verification of model-serving reliability.

The final CatBoost Regressor achieved an R^2 score of 0.682, an MAE of \$2.64 per hour, and an RMSE of 5.40. Five-Fold Cross-Validation produced an average MAE of 2.64 with a standard deviation of ± 0.048 , demonstrating stable model performance across different data subsets. Linear Regression was used as a baseline model for comparison.

14. Innovation and Added Value

SkillBridge introduces a data-driven decision intelligence framework that goes beyond conventional freelance-market reporting. Its main innovation lies in integrating Business Intelligence, skill-gap analytics, opportunity assessment, and Machine Learning into a unified solution that converts complex market data into actionable strategic insights.

Rather than analyzing demand, skills, competition, and earnings as isolated indicators, SkillBridge establishes analytical relationships between them to provide a multidimensional view of the freelance ecosystem. The Diamond Skills Finder identifies strategically valuable skills by evaluating market demand, available talent supply, skill gaps, competition intensity, and earning potential, enabling stakeholders to distinguish between highly competitive skills and emerging high-opportunity skills.

The solution further extends descriptive and diagnostic analytics with predictive intelligence through a Machine Learning–based Salary Predictor. The implemented CatBoost Regressor estimates expected hourly market rates using relevant job and market characteristics, providing an additional evidence-based layer for evaluating the economic value of freelance opportunities.

At the strategic level, SkillBridge creates a bridge between labor-market intelligence and human-capital development. Its insights can support governmental entities, educational institutions, universities, and technology training providers in identifying priority digital skills, aligning curricula and training programs with evolving market demand, and directing skill-development initiatives toward areas with stronger employment and earning potential.

The added value of SkillBridge therefore lies not in a single dashboard or predictive model, but in the integration of market intelligence, skill intelligence, predictive analytics, and strategic decision support within one coherent framework—transforming freelance-market data from descriptive information into actionable intelligence for workforce development and future-ready skills planning.

15. Competitive Analysis & Differentiators

1. Existing Solutions & Market Gaps

Existing platforms and market tools generally fall into two distinct silos:

Traditional Freelance Marketplaces (e.g., Upwork, Mostaq, Khamsat): Provide transactional job listings and talent sourcing but lack transparent wage benchmarking, localized skill analytics, and objective guidance for entry-level freelancers to navigate pricing or skill saturation.

General EdTech & E-Learning Platforms (e.g., Coursera, Udacity, local coding academies): Offer technical training and skill acquisition courses but operate without a direct, empirical feedback loop connecting their curriculum ROI to active, real-time freelance market demand and candidate acceptance rates.

2. The SkillBridge Advantage (Competitive Edge)

SkillBridge bridges this structural gap by integrating educational initiatives, labor market data, and predictive machine learning into a single end-to-end analytical ecosystem.

Dimension / Feature	Traditional Job Platforms	General EdTech Platforms	SkillBridge Platform
Multi-Platform Aggregation	Isolated (Single platform)	None	Integrated (Upwork, Mostaq, Khamsat, Kaggle)
Educational ROI Alignment	None	Limited to course completion	Direct K-12 & Training ROI vs. Market Acceptance
Predictive Pricing Engine	Basic / Manual budget ranges	None	CatBoost Machine Learning ($R^2 = 0.682$, MAE = \$2.64/hr)
Skill-Gap Transparency	Implicit / Saturated listings	Static skill paths	Dynamic Skill Demand-Supply Mapping & Rate Boundaries
Target Decision Support	Freelancer/Client only	Learner only	Multi-Stakeholder (MCIT, Ministries, Academies, Talents)

3. Key Differentiators (Why SkillBridge Stands Out)

Empirical Educational Feedback Loop: Unlike traditional platforms, SkillBridge quantifies the direct impact of pre-university (K-12) technical training on employment outcomes, providing decision-makers (e.g., MCIT) with actionable metrics (Post_Acceptance_Rate, hourly earnings uplift) to optimize educational grants.

Machine Learning-Driven Wage Standardization: Features a dedicated CatBoost Regressor pricing model trained on 100,000+ job records. This prevents price undercutting for beginners while establishing fair, objective hourly rate boundaries (Estimated_Hourly_Rate) for both hourly and fixed-price contracts.

Interdisciplinary & Strategic Vision: Designed to serve a multi-stakeholder ecosystem—aligning national digital transformation strategies with private EdTech pipelines and individual talent empowerment.

16. Team Roles and Responsibilities

Define the role of each team member. Roles should be adjusted based on the project track and team size.

Team Member	Role	Responsibilities	Tasks
Alaa Hussien Mohamed	Data Engineer & Web Scraping Specialist, AI Specialist	Implementation, coding, and testing.	Scraping, EDA, ML, Documentation.
Rasha Wael Ramadan	Developer / Engineer, Business Intelligence (BI) & Dashboard Spe- cialist	Data preparation, model building, analysis, or pipelines.	Dashboards, Docu- mentation.
Fatma Ibrahim Morsy	Dashboard Specialist UI/UX / Frontend	Interface design and frontend development.	Dashboards, Inter- face.
Esraa Metwally Abo Elnasr	Project Coordinator Dashboard Specialist	Planning, coordina- tion, documentation, and communication.	Dashboards, Presentation.

17. Ethical, Security, and Privacy Considerations

SkillBridge follows responsible data and AI practices to ensure that freelance-market data, analytical results, and predictive outputs are handled ethically, securely, and transparently. The main considerations include:

- **Protecting Data Privacy:** Avoiding the exposure of sensitive or unnecessary personal information in datasets, dashboards, reports, and API outputs.
- **Ethical Data Usage:** Using freelance-market data only for analytical, educational, research, and decision-support purposes.
- **Data Minimization:** Processing only the information required for market analysis, skill-gap assessment, and Machine Learning prediction.
- **AI Bias Awareness:** Monitoring potential bias across countries, job categories, platforms, skills, and experience levels.
- **Data Leakage Prevention:** Excluding post-publication variables such as Applicants_Count, Proposals_Count, Hire_Rate, and Job_Status from the Salary Predictor.
- **Prediction Transparency:** Presenting Salary Predictor results as estimated market rates rather than guaranteed salaries or income.
- **Secure API Integration:** Applying input validation, controlled model access, error handling, and secure handling of prediction requests.
- **Data and Model Integrity:** Protecting datasets, trained models, project files, and source code from unauthorized modification or misuse.
- **Human-Centered Decision Support:** Using SkillBridge insights to support, rather than replace, human decisions in education, training, and workforce planning.

18. Expected Impact

SkillBridge is expected to create measurable value across workforce development, education, training, market intelligence, and individual career decision-making. The following impact areas represent the most relevant outcomes of the proposed solution.

Impact Area	Example Questions
User Impact	Helps students, graduates, and freelancers understand market demand, identify promising skills, evaluate competition, explore opportunities, and estimate expected hourly rates more effectively
Business / Organizational Impact	Supports training providers, universities, and public-sector stakeholders in making more evidence-based decisions regarding skills development, training priorities, and workforce planning.
Technical Impact	Demonstrates the integration of Power BI, DAX, Machine Learning, and API services within a unified decision-support framework for freelance-market intelligence.
Social Impact	Supports digital inclusion and employability by helping individuals better understand the skills required to access freelance and remote-work opportunities.
Economic Impact	Can improve the allocation of training resources, reduce misalignment between learning and market demand, and direct attention toward skills with stronger employment and earning potential.
Educational Impact	Helps educational institutions and technology training providers align curricula and training programs with real market demand, skill gaps, and emerging digital opportunities.

19. Conclusion

• Strategic Vision & Core Purpose

SkillBridge represents an integrated data-driven framework developed to transform complex freelance-market data into actionable intelligence for skills development, opportunity assessment, and strategic workforce decision-making. The project addresses a fundamental challenge in the digital labor market: the disconnect between evolving market demand, available skills, educational and training priorities, and sustainable earning opportunities.

• End-to-End Technical & BI Implementation

The project successfully established an end-to-end analytical workflow encompassing data preparation, integration, validation, relational data modeling, KPI development, DAX-based analysis, and interactive Business Intelligence reporting. Through its Market Intelligence, Skill Gap Intelligence, Diamond Skills Finder, and Opportunity Intelligence capabilities, SkillBridge enables multidimensional analysis of market demand, talent supply, skill shortages, competition intensity, hiring activity, and earning potential.

• Machine Learning & Predictive Performance

Predictive analytics further strengthens the framework through the Machine Learning-based Salary Predictor. The final CatBoost Regressor, trained on 109,470 valid job records, achieved an R^2 of 0.682, an MAE of \$2.64 per hour, and an RMSE of 5.40. The use of Five-Fold Cross-Validation and Data Leakage prevention measures provides additional evidence of model stability and methodological reliability.

• Ecosystem & Stakeholder Impact

Beyond its technical implementation, the strategic contribution of SkillBridge lies in its ability to connect labor-market intelligence with human-capital development. The generated insights can support governmental entities, educational institutions, universities, and technology training providers in identifying priority digital skills, improving curriculum and training alignment, and directing skills-development initiatives toward areas of measurable market opportunity. Students, graduates, and freelancers can similarly benefit from greater transparency regarding skill demand, competition, and expected earning potential.

• Final Conclusion

In conclusion, SkillBridge demonstrates that the integration of Business Intelligence and Machine Learning can move freelance-market analysis beyond descriptive reporting toward evidence-based decision intelligence. By establishing a measurable link between education, skills, market demand, opportunities, and earning potential, the project provides a strong foundation for more responsive digital-skills planning and future workforce development.

20. Future Work

- The future development of SkillBridge aims to transform the current analytical solution into a comprehensive and personalized digital career-support platform that connects users directly with suitable job opportunities, learning resources, government training programs, and practical freelancing guidance based on their individual skills and career goals. Users will be able to enter their current skills into the platform, allowing SkillBridge to analyze their skill profiles and recommend freelance and remote job opportunities that best match their capabilities. The platform can also aggregate and organize relevant job links from multiple freelance and remote-work platforms within a centralized interface, reducing the time and effort required to search across different sources. A skill-based recommendation engine can compare each user's skills with job requirements, calculate the level of compatibility, and rank available opportunities according to their relevance to the user's profile.
- In addition to job recommendations, SkillBridge can provide personalized skill-gap analysis by identifying the skills a user currently possesses and comparing them with the skills required for selected jobs or career paths. Based on these gaps, the platform can recommend the most important skills to develop and create personalized learning pathways that connect users with relevant courses, training opportunities, and professional development resources. SkillBridge can also recommend suitable scholarships, training programs, and digital-skills initiatives provided by the Ministry of Communications and Information Technology (MCIT) and other governmental, educational, and training institutions according to the user's current skills, interests, career direction, and identified development needs.
- Another important future component is the development of a dedicated Freelancing Knowledge and Guidance Center for individuals entering the freelance market for the first time. This section can include practical digital guidebooks and educational resources explaining how freelance platforms operate, how to create a professional profile, search and apply for suitable projects, write effective proposals, communicate professionally with clients, determine appropriate pricing, manage projects, deliver work successfully, build strong ratings, and follow professional practices on different freelance platforms. Platform-specific guidance can also be provided to help new freelancers understand registration procedures, platform rules, bidding systems, payment processes, communication methods, and best practices.
- Through these developments, SkillBridge can evolve from a Business Intelligence and Machine Learning solution into an integrated digital career ecosystem that connects Skills → Skill Gaps → Learning Opportunities → Government Programs → Suitable Jobs → Freelancing Guidance → Career Growth. This future vision would enable SkillBridge to support users throughout the complete journey from understanding their current capabilities and developing market-relevant skills to discovering suitable opportunities and participating successfully in the freelance and digital labor market.

21. Strategic Recommendations

Based on empirical insights and predictive modeling from SkillBridge, the team proposes actionable recommendations for key stakeholders:

1. Educational Policy & Curriculum Reform (MCIT & Ministries of Education)

Market-Aligned Curricula: Periodically update K-12 and university tech modules based on real-time market deficits, prioritizing high-demand skills (e.g., Cloud Architecture, AI/ML, Data Engineering).

Peer-Led Practical Training: Deploy graduates of national initiatives (e.g., DiGiLi-ANS) to teach coding in schools, shifting the educational focus from theoretical concepts to hands-on, project-based learning.

Empirical Grant Allocation: Direct government IT training funds toward high-impact skills using SkillBridge's educational ROI metrics (Post_Acceptance_Rate and wage uplift).

2. Talent Empowerment & Private EdTech

Data-Driven Upskilling & Pricing: Guide talents using the CatBoost Salary Predictor to set fair hourly rates and transition from saturated basic tasks to high-margin technical combinations.

Integrated Skill Bundling: Design training pipelines that merge technical skills with freelance survival skills (proposal writing, contract negotiation, and portfolio building).

3. Governance & Regulatory Framework

Freelancer Legal Protection: Formulate standardized digital contract templates and legal guidelines to safeguard local freelancers against payment defaults on global platforms.

National Digital Talent Registry: Establish a unified national registry to streamline cross-border foreign currency remittances and tax frameworks for entry-level digital talents.

22. References / External Resources (Optional)

The following external resources, datasets, libraries, frameworks, and official technical documentation were used to support the design, development, analysis, Machine Learning implementation, and API integration of the SkillBridge project:

1. Microsoft. Power BI Documentation. Microsoft Learn — Reference for data modeling, dashboard development, visualization, and Business Intelligence implementation.
2. Microsoft. Data Analysis Expressions (DAX) Reference. Microsoft Learn — Reference for developing measures, KPIs, and analytical calculations used in SkillBridge.
3. Microsoft. Power Query Documentation. Microsoft Learn — Reference for data cleaning, transformation, preparation, and integration.
4. Python Software Foundation. Python Documentation — Reference for data processing, feature engineering, Machine Learning development, and API implementation.
5. Pandas Development Team. Pandas Documentation — Used for data manipulation, cleaning, transformation, and preparation.
6. NumPy Developers. NumPy Documentation — Used for numerical operations and data processing within the Machine Learning workflow.
7. Scikit-learn Developers. Scikit-learn User Guide and Documentation — Reference for data preprocessing, baseline modeling, model evaluation, and cross-validation.
8. CatBoost Developers. CatBoost Documentation — Reference for developing and evaluating the CatBoost Regression model used in the SkillBridge Salary Predictor.
9. FastAPI. FastAPI Documentation — Reference for developing the API component used to integrate and serve the Machine Learning prediction model.
10. Pydantic. Pydantic Documentation — Reference for structured data validation and prediction-request handling within the API.
11. Joblib Developers. Joblib Documentation — Used for saving, loading, and managing the trained Machine Learning model.
12. Uvicorn. Uvicorn Documentation — Reference for running and testing the FastAPI-based model-serving application.
13. Ministry of Communications and Information Technology (MCIT), Egypt. Digital Skills Development and Training Initiatives — Contextual reference for digital-skills development, training initiatives, and workforce-development objectives.
14. SkillBridge Freelance Market Dataset — Project dataset containing freelance jobs, skills, freelancers, clients, platforms, countries, job categories, job-skill relationships, and educational initiative data used for Business Intelligence analysis and Machine Learning development.
15. SkillBridge Project Team. (2026). SkillBridge Project Documentation and Technical Implementation — Power BI data model, DAX measures and KPIs, interactive dashboards, Skill Gap Intelligence, Diamond Skills Finder, Opportunity Intelligence, Machine Learning Salary Predictor, and API integration developed as part of the Nano Degree Capstone Project.

Appendices

Appendix A: Screenshots / Prototype Design

Add optional website pages, mobile screens, dashboard designs, system interface mockups, model output examples, security alert examples, or data pipeline screenshots.



Figure 2 SkillBridge End-to-End Project Development Journey & Implementation Roadmap

```

import pandas as pd
import requests
from bs4 import BeautifulSoup

[1]

# تحديد الرابط الرئيسي لوظائف تحليل البيانات على موقع مستقل
url = "https://mostaql.com/projects?category=development&keyword=data"

headers = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,"
        " like Gecko) Chrome/115.0.0.0 Safari/537.36"
    )
}

[2]

# 2. إرسال الطلب وجلب محتوى الصفحة
response = requests.get(url, headers=headers)

if response.status_code == 200:
    soup = BeautifulSoup(response.text, "html.parser")
    projects = soup.find_all("tr", class_="project-row")

    scraped_data = []

    # 3. استخراج البيانات من كل مشروع
    for project in projects:
        # المسمى الوظيفي
        title_tag = project.find("h2", class_="project-title")
        title = title_tag.text.strip() if title_tag else "N/A"

        # رابط المشروع
        link = (
            title_tag.find("a")["href"]
            if title_tag and title_tag.find("a")
            else "N/A"
        )

        # الميزانية وحالة المشروع
        budget_tag = project.find("span", class_="project-budget")
        budget = budget_tag.text.strip() if budget_tag else "N/A"

        # عدد العروض المقدمة
        proposals_tag = project.find("span", class_="project-proposals")
        proposals = proposals_tag.text.strip() if proposals_tag else "N/A"

        scraped_data.append({
            "Platform": "Mostaql",
            "Job Title": title,
            "Budget": budget,
            "Proposals_Count": proposals,
            "Job_URL": link,
        })

    # 4. وحفظها في DataFrame تحويل البيانات لـ CSV
    df = pd.DataFrame(scraped_data)
    df.to_csv("mostaql_scraped_data.csv", index=False, encoding="utf-8-sig")
    print("أتم سحب البيانات وحفظها بنجاح")
else:
    print(f"نشل الاتصال بالموقع. رمز الحالة {response.status_code}")

[3]

... أتم سحب البيانات وحفظها بنجاح

import os

print("مسار الملف على جهازك هو:")
print(os.path.abspath("mostaql_scraped_data.csv"))

[4]

... مسار الملف على جهازك هو:
c:\Users\A-FF-I4-D10\AppData\Local\Programs\Microsoft VS Code\mostaql_scraped_data.csv

```

Figure 3 Automated Web Scraping Script for Freelance Data Extraction

Freelance Jobs — EDA & Data Cleaning Project

Project Objective

This notebook performs a complete **Exploratory Data Analysis (EDA)** and **Data Cleaning** workflow on a deliberately dirty freelance jobs dataset.

The workflow covers:

- Dataset inspection and structure
- Missing values and NULL-like values
- Duplicate records
- Data type validation
- Date quality checks
- Numeric outlier detection
- Text quality checks
- Data cleaning and validation
- Exporting the cleaned dataset
- Final visualizations and business insights

Important: The dirty dataset was created by appending corrupted copies of original records. Therefore, the original clean version can be recovered safely by keeping the **first occurrence of each Job_ID**.

1. Import Libraries

We start by importing the Python libraries required for data manipulation, analysis, and visualization.

```
[1] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

pd.set_option("display.max_columns", None)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")

plt.rcParams["figure.figsize"] = (10, 5)
plt.rcParams["axes.titlesize"] = 14
```

3. Initial Data Inspection

Before cleaning, we inspect the structure of the data, column names, data types, and sample records.

```
[ ] display(df.head())

[ ] print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

print("\nColumn Names:")
for col in df.columns:
    print("-", col)

[ ] print("Data Types:")
display(df.dtypes.to_frame("Data Type"))

[ ] print("General Dataset Information:")
df.info()
```

4. Statistical Summary

We inspect descriptive statistics for numeric columns to identify unusual ranges, possible outliers, and invalid values.

```
[ ] display(df.describe(include="number").T)
```

5. Missing Values & NULL-like Values

Figure 4 Exploratory Data Analysis (EDA) & Data Cleaning Pipeline in Python

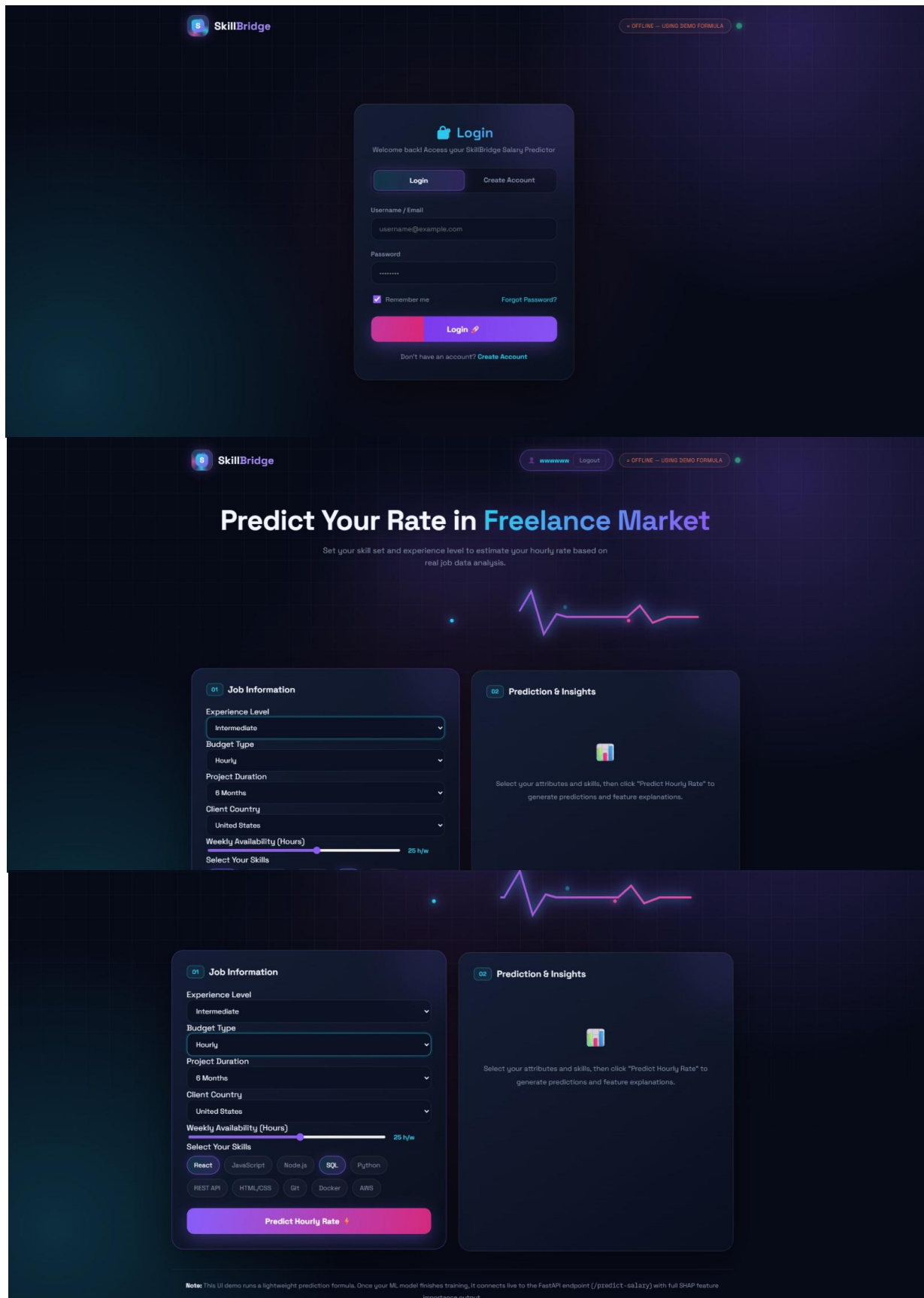


Figure 5 Machine Learning Salary Predictor Interface & Feature Explainability UI



Explore the Dashboard >



Market Overview

Explore freelance market trends and key performance indicators



Skill Gap Intelligence

Analyze skill demand, supply and market gaps



Diamond Skills Finder

Discover high-demand, high-value skills with lower competition



Opportunity Intelligence

Analyze jobs, platforms, competition, and earning opportunities



Learning & Career Pathway

Connect skills with learning opportunities and career development



Egypt 2030 Vision

Explore digital workforce insights supporting Egypt's future



About SkillBridge

Bridging skills, market demand, and opportunities through data-driven intelligence for a future-ready workforce.



Freelance Opportunity Intelligence

Turning Market Data into Smarter Career Opportunities

Total Jobs

111K

Total Clients

12K

Skills Tracked

22K

Hire Rate

37%

Applicants / Job

28.5

Avg. Hourly Rate

\$11.81

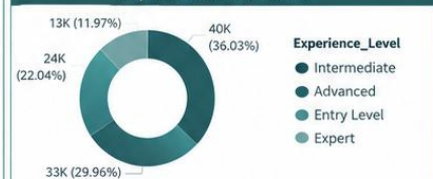
Market Growth & Opportunity Trend



Top Job Categories



Experience Level Demand



Platform Opportunity Overview

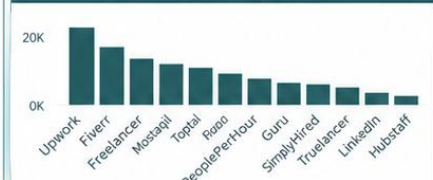


Figure 6 SkillBridge Interactive Power BI Dashboard Suite & Market Opportunity Overview

Appendix B:

Additional Diagrams

Add optional use case diagrams, flowcharts, database schemas, model architecture, data pipelines, network/security architecture, or deployment diagrams.

Appendix C: Useful Links

Item	Link
GitHub Repository	
Demo Video	
Prototype Link	
Dataset Link	
Dashboard Link	